

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I G. Gowthami/192524285 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G. Gowthami

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments

- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

ANSWERS

1. AI-Powered Drone Delivery (Greedy Best-First Search & A*)

i. Greedy Best-First Search

Selects the path with the smallest heuristic value (estimated distance).

Quickly reaches the destination.

Ignores the actual travel cost.

May choose unsafe or blocked routes.

Strengths:

Fast decision-making.

Low memory usage.

Suitable for urgent situations.

Weaknesses:

Does not guarantee the shortest path.

Easily affected by incorrect heuristic values.

Performs poorly when paths suddenly become blocked.

ii. A* Search

Uses $f(n) = g(n) + h(n)$.

$g(n)$: Actual travel cost.

$h(n)$: Estimated remaining distance.

Finds the safest and lowest-cost path.

iii. Impact of Heuristic Accuracy

Accurate heuristic → Faster and better solutions.

Poor heuristic → More nodes explored and slower performance.

A* remains optimal with an admissible heuristic.

Greedy Search is more sensitive to heuristic errors.

iv. Effect of Dynamic Changes

Blocked roads require replanning.

Greedy may repeatedly choose blocked paths.

A* updates costs and searches alternative routes.

v. Recommendation

A Search* is the best choice because it:

Produces optimal routes.

Considers both distance and travel cost.

Adapts better to changing environments.

Provides safer navigation.

2. Smart Traffic Signal Optimization

Optimization Problem

Objective: Minimize waiting time, fuel consumption, and congestion.

Traditional search is inefficient because:

Huge number of signal combinations.

Traffic changes continuously.

Search space is very large.

i. Hill Climbing

Starts with one solution.

Moves to a better neighboring solution.

Stops when no better neighbor exists.

Limitations

Gets stuck in local maxima.

Cannot escape plateaus.

Misses the global optimum.

ii. Local Maxima, Plateaus and Ridges

Local Maximum: Good but not the best traffic pattern.

Plateau: Many solutions have the same performance.

Ridge: Best path requires multiple changes together.

iii. Simulated Annealing & Genetic Algorithm

Simulated Annealing

Sometimes accepts worse solutions.

Escapes local maxima.

Finds better overall solutions.

Genetic Algorithm

Uses selection, crossover and mutation.

Explores many solutions simultaneously.

Produces high-quality signal timings.

iv. Recommendation

Genetic Algorithm

Highly scalable.

Adapts to changing traffic.

Finds near-optimal solutions efficiently.

3. Mars Rover (Online Search Agent)

i. Why Online Search?

Complete map is unavailable.

Environment is unknown.

Rover learns while moving.

Decisions are made in real time.

ii. Challenges

Partial observability.

Hidden obstacles.

Limited sensor range.

Dynamic environment.

iii. Internal Model

The rover:

Stores explored locations.

Updates obstacle information.

Marks safe and unsafe paths.

Improves navigation continuously.

iv. Online vs Other Search

Online Search

Learns while exploring.

Suitable for unknown environments.

Uninformed Search

No heuristic.

Explores many unnecessary paths.

Informed Search

Uses heuristic.

Works best when map information exists.

v. Recommendation

Online Search Agent

Handles uncertainty.

Continuously updates knowledge.

Safe and adaptable.

Ideal for Mars exploration.

4. University Exam Timetable (CSP)

CSP Formulation

Variables

Exams

Domain

Available dates, time slots and halls

Constraints

No overlapping exams.

Limited halls.

Faculty availability.

Subject order.

Special accommodations.

ii. Backtracking Search

Assign one exam.

Check constraints.

If conflict occurs, backtrack.

Continue until a valid timetable is found.

iii. Constraint Propagation (Forward Checking)

Removes invalid future choices.

Detects conflicts early.

Reduces search space.

Improves efficiency.

iv. Real-World Challenges

Large number of students.

Many constraints.

Limited resources.

Last-minute schedule changes.

Best Strategy

Backtracking with Forward Checking because it efficiently handles complex constraints and scales well.

5. Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax

Models the game as a decision tree.

AI maximizes its score.

Opponent minimizes the AI's score.

Chooses the best possible move.

ii. Computational Challenges

Huge game tree.

High memory usage.

Long computation time.

Difficult for real-time games.

iii. Alpha-Beta Pruning

Removes unnecessary branches.

Produces the same optimal result as Minimax.

Explores fewer nodes.

Increases search speed.

Enables deeper search.

iv. Evaluation Function

Factors:

Player score.

Resources available.

Army strength.

Map control.

Distance to objectives.

Health of units.

v. Recommendation

Use Minimax with Alpha-Beta Pruning because it:

Reduces computation.

Maintains optimal decisions.

Works efficiently in real-time strategy games.

Balances speed and decision quality.

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand